

Securing Your Cyber Security: What is Hacking?

Aditya Anand

1 Introduction

People often hear stories of AI taking over the world, or of malicious software posing a threat to your personal information. However, the individuals behind such cyber attacks are sometimes overlooked. With the internet being such a vast space, hackers seek the opportunity to take advantage of any opening they can find. Here are some of the most common techniques hackers use to steal your information.

2 Five Common Strategies Used by Hackers

2.1 Malware

It is important to understand how hackers attack their virtual prey. One of five strategies is usually used, with the most common being the utilization of malware. Malware is short for malicious software, and it consists of harmful code or programs. These programs are attached to websites or emails, and when downloaded, can be used to access personal files and steal data, such as login credentials. Malware can enter the computer in multiple ways, including downloads from websites and removable storage hard drives.

2.2 Social Engineering

Another method commonly used is called social engineering. This involves manipulating “individuals to unknowingly disclose confidential information or personal data, which hackers use for fraudulent purposes.”[1] After convincing the victims that the hackers are a legitimate and trustable source, the hackers can also send links and make use of malware. Common examples of social engineering include scam emails promising free rewards or fake organizations asking for contact information.

2.3 Hacking

Hackers may also use a strategy simply known as “hacking”. As established, hacking is much broader than one method, but this method is often the first step in a cyberattack and involves the use of vulnerabilities in the operating system, such as the lack of encryption, as well as the physical device. This allows hackers to gain unauthorized access to programs, data, networks, or computers, which they use to steal money, make unauthorized purchases, and sell personal information to third parties.

2.4 Credential Compromise

While many methods focus on downloading software to gain access to private information, “credential compromise” targets password managers, such as the Google Password Manager. This form of hacking is completed through an application that automates entry by trying thousands of possible passwords per minute. This allows hackers to view valuable information such as social security numbers, bank accounts, and passwords. Once hackers have access to your password manager, they are able to commit identity theft, transfer money, and access private information.

2.5 Distributed Denial of Service

A fifth method is used more commonly for industrial and political gain. “DDos” (Distributed Denial of Service), is the process of overwhelming a website with more traffic than the server or network can accommodate. Traffic is the number of web users who visit a website, and higher traffic is typically beneficial for the business. However, an unmediated rise in traffic leaves the targeted website inoperable, which can have a variety of impacts. DDos may be used to cause an organization to lose money or popularity, benefitting rival organizations.

3 Conclusion

Overall, there are a variety of methods that hackers can use to steal personal information, money, and passwords. However, there are also multiple ways to avoid such cyber attacks. Downloading software updates, using virtual private networks (VPNs), and avoiding suspicious links and ads are all effective forms of protection. While AI could pose a threat to humanity, it is the people behind these attacks that we must be aware of and defend against.

References

Fortinet. 2022. “What Is Hacking? Types of Hacking & More.” Fortinet. 2022.

<https://www.fortinet.com/resources/cyberglossary/what-is-hacking>.

“Understanding Hacking Strategies (How Hackers Hack) – Stage2Data.” n.d.

<https://stage2data.com/understanding-hacking-strategies-how-hackers-hack/#:~:text=How%20do%20hackers%20use%20it>.